cent climate change conference (COP26) held in Glasgow. The most important of the commitments made is that India will meet 50 per cent of its energy requirements from renewable sources by 2030. This is big—no question about it. The fact is energy system transformation is what will drive mitigation in the world. In effect, what India is saying is that its future energy infrastructure will be 'green'—it will not give up coal, but it will drastically reduce its dependence on this black gold, which is indicted as the key cause of climate change emissions.

Let's look at the numbers. Currently, India's dependence on coal to meet its energy needs is near absolute—in 2019, coal-based thermal power was 63 per cent of the installed capacity and met 80 per cent of the country's energy needs. But if this plan has to go through, then by 2030 the country will have to turn this around. According to the Central Electricity Authority, dependence on coal-based energy would go down to about 56 per cent and the rest would be from renewable energy. To achieve this target, India would need to increase its renewable portfolio from 100 Giga Watt (GW) currently to between 650-700 GW by 2030, while coal-based energy would be roughly 280 GW—a marginal increase from now. This sounds daunting, as it should. But it is also feasible, if we plan it deliberately and with imagination.

The fact is, we need a reinvention of our energy system to meet critical livelihood and economic needs of millions of our people. Today, India's energy system is so broken that it takes the cheapest energy generation—coal-based thermal power—and supplies it so that it becomes, arguably, the most expensive in the world. The reasons are manifold; from the perverse subsidy systems that end up *not* benefitting the poor users and farmers, in whose name it is given, to the huge leakages and inefficiencies in the power distribution companies (discoms). But the bottom-line is that India continues to be debilitated by energy poverty as millions remain in the dark and women continue to cook on biomass-based stoves, which are bad for their health. On the other hand, the richer consumers opt out of the same broken energy system to use even dirtier sources, from diesel for generators to coal and petcoke for boilers and captive power plants. All this has deadly health impacts as coal use grows in both highly polluting power plants and individual boilers. This is not economical or sustainable. It has to change.

This is where the new energy systems can play a role. But it requires investment, not just in building infrastructure, but in rebuilding the distribution and supply network to be clean and efficient. The real challenge is not building infrastructure but ensuring even the poorest can pay the cost of this 'clean' energy. This is really where the nub of the problem is. We need to look beyond renewable to see how this clean energy can be affordable so that growth is inclusive. This is the opportunity to use climate finance so that it pays not just for setting up clean energy plants but pays the difference in the cost that it would take to make this energy accessible to all.

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So, what needs to be done in 2022? First, we need a plan for climate change action in the country. This has to be 'business as unusual' and has to work out the options for emission reduction in each sector of India's economy and how we will benchmark change. We have always been reluctant to do this, given that it would bind us to emission reduction targets globally. But we have to bite this bullet. We know we have options today to reinvent growth without pollution. We will build our energy, building, industry and city mobility infrastructure. So, the opportunity is to redesign to make it more resource efficient, more affordable and, as a result, more sustainable. The plan needs numbers and a roadmap on how we will get to 2030 and what it will cost—this funding requirement is important as climate change finance has to be part of this transformation.

The second agenda for 2022 is to work on a strategy to address the horrendous impacts of climate change—from flooding because of extreme rain and drought because of scarce rain to increased frequency of tropical cyclones or other weather changes. We need intervention strategies to address these changes and fast. For instance, the farmers need increased support for crop losses and, for this, the government must rework the insurance scheme to meet this challenge. Then we need to ensure that every time it rains, our cities do not go underwater—this needs the urban renewal programme to focus on protection of lakes, ponds and green areas in settlements. The biggest gamechanger would be in building resilience against these repeated shocks by investing in water conservation and green livelihoods in rural India. I could go on. But the real change in 2022 is that we get serious about this existential threat. ■

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